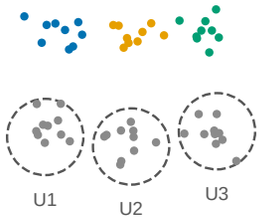
Gene-based baselines: standard single-cell tools + connectome-guided matching

1 Cells



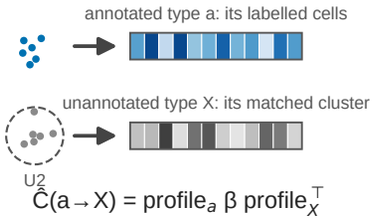
annotated cells keep their label;
unannotated cells (grey)
have no cell type

2 Clustering



Seurat / Scanpy / Cell Ranger group
unannotated cells into clusters
U1, U2, … (type unknown)

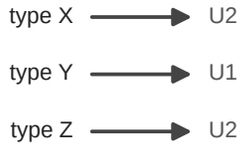
3 Profiles + gene rule



profile = mean log-expression of the
unit's cells (2,000 HVGs); β fitted on
ANNOTATED training types (~65 / 16 / 9)

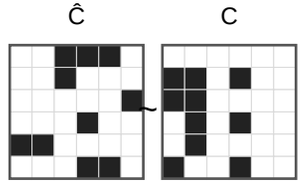
uses held-out C

4 Matching



each unannotated test type gets
ONE cluster, chosen to maximize
r with the held-out connectome

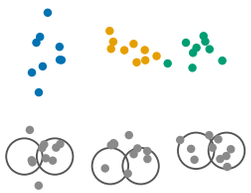
5 Predict + score



Ĉ for every test pair from their
profiles; score r(Ĉ, C) on test types

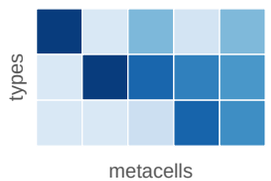
ConnectionMiner

1 Metacells



cells aggregated into metacells
within clusters; annotated
metacells pinned to their type

2 Soft assignment P



every metacell spreads over its
candidate types (transport plan);
annotated stay with their type

3 Profiles + gene rule β



$\hat{C} = E\ \beta\ E^{T} = P\ G\ \beta\ G^{T}\ P^{T}$

profile $E_t = P_t G$: detection of the metacells
weighted by P; β learned jointly with P on
ALL training types (~590 / 680 / 580)

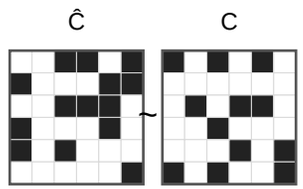
uses held-out C

4 Test types

soft, metacell-level
(many metacells per type)

β frozen; P solved for the test
types by Sinkhorn-projected
mirror descent on the held-out C

5 Predict + score



$\hat{C} = P\ G\ \beta\ G^{T}\ P^{T}$ on test types
score r(Ĉ, C)

Red frame: the step uses the held-out connectome. In the baselines, a rule fitted on shuffled profiles scores almost the same, so their r comes mostly from matching.